# Ir. Olivier **NDINGA OBA**

## Software Developer

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## PROFILE

MSc graduate in Computer Science & Engineering with a strong military background (Royal Military Academy). I combine analytical rigor with advanced programming skills (C++, Python). Passionate about designing high-performance software and complex systems, I have hands-on experience in algorithm optimization, machine learning, and data analysis. I am currently seeking a Software Developer position to build robust, scalable, and innovative software solutions.

## EXPERIENCE

2024 - 2025 | Master's thesis on behavioral imitation in robotic swarms under the supervision of the IRIDIA laboratory

2022 - 2024 | **Software Developer & Data Analyst** Voisin d'énergiE & SNCB (Société Nationale des Chemins de fer Belges)

2020 - 2022 | Student Equipment Manager

2018 - 2022 | Military Training in an International Context

## CERTIFICATIONS

Oct 2025 IBM Data Engineering Essentials

Oct 2025 IBM Introduction to Containers w/ Docker, Kubernetes

Sep 2025 Tableau, Networks & Time Series Data Visualization

Aug 2025 Reinforcement Learning Specialization

Jul 2026 | Google AI Professional

Aug 2026 | Foundations of Project Management

Aug 2026 | Git and GitHub

## COMPETITIONS

Nov 2025 Finalist, Future of IT Leaders in Belgium: Data & AI Challenge

Nov 2025 Finalist, Huawei France Tech Arena: The Data

- Designed a traffic allocation algorithm for dynamic UAV 6G networks to maximize throughput and minimize latency

## EDUCATION

**Master of science in Computer Science and Engineering** (Distinction)

**Bachelor of Science in Engineering Science**

## SKILLS

- Programming Languages: Advanced in C/C++, Python. Proficient in SQL, basic JavaScript
- Technologies & Tools: Git/GitHub, Docker, Linux, ROS/ROS2, Vercel
- Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, Agile/Scrum

## LANGUAGES

- French (native)
- English (professional)

## PERSONALITY

- Time Management, Autonomy Initiative, Critical Thinking, Ethics & Professionalism

# PROJECTS

**MASTER THESIS**

### IRIDIA Swarm robotics

Conducted research on behavioral imitation in robotic swarms, demonstrating the value of full field data for accurate behavior characterization, creating novel metrics for effective cross-architecture transfer between neural and rule-based controllers, and extending the approach to biological systems by modeling fish schooling.

Skills used - **Mathematical Modeling, TDA, Controller design, Statistics, Experimental design, benchmarking, Python, C++, Argos, GUDHI**

**DATA MINING**

### SNCB Trains anomaly detection

Engineered a scalable Python framework for automated anomaly detection in large-scale train datasets.

Skills used - **R, Python, Data Pipelines, Scrum, Agile Team Leadership**

**STATISTICAL OF ML**

### Urban Air Pollution

This project addresses the Urban Air Pollution Challenge Zindi competition by implementing and assessing various supervised learning algorithms along with different feature selection methods for the associated regression task.

Skills used - **R, Python, Machine learning**

**OPERATIONAL RESEARCH**

### Genetic agriculture optimization

Built an optimization engine in Python using Genetic Algorithms to maximize agricultural site selection based on budget, land use, and proximity constraints

Skills used - **Python, Algorithm Design, Optimization (Promethee)**

**TECHNIQUES OF AI**

### Predicting electricity consumption with LSTM

Designed and developed a complete LSTM neural network entirely from scratch in C++ to predict energy consumption for the "Voisin d'Énergie" community network. Demonstrated strong memory management and Object-Oriented Programming skills.

Skills used - **C++, Object-Oriented Programming, Machine Learning**